

Linear Equations

Matrix Form $AX=B$

Solution Vector X

Systems of Linear Equations & Determinants

MATHEMATICS FOR AI/ML

Equations → Matrix Form → Determinants → Solutions

1

Linear Equations

Constraints on unknowns expressed algebraically

2

Matrix Form $AX = B$

Compact representation enabling powerful tools

3

Determinants

Scalar health-check on the coefficient matrix

4

Solutions

Unique, infinite, or none — determined by $\det(A)$

Why Do We Need Linear Equations?

Real-World Motivation

- 🔗 **Pricing problems:** Two products, two unknowns — find individual prices from total bills
- 🔗 **Resource allocation:** Distribute limited resources across tasks with constraints
- 🔗 **ML model parameters:** Fitting a line $y = mx + c$ requires solving for m and c
- 🔗 **Engineering & physics:** Circuit analysis, force balance, fluid flow — all reduce to linear systems

A Simple Example

You buy apples and bananas. Together they cost ₹10. Two apples and one banana cost ₹15.

Equation 1

$$x + y = 10$$

Total cost constraint

Equation 2

$$2x + y = 15$$

Weighted purchase constraint

What are x and y ? → This is a **system of linear equations**.

- 📌 Each equation is a constraint. Together, they narrow down the answer to a unique point.

What Is a System of Linear Equations?

Key Vocabulary

Linear Equation

Each variable appears with power 1 — no x^2 , no xy

Variable

The unknown quantity we want to find ($x_1, x_2, \dots, x_n$)

Coefficient

The number multiplying a variable ($a_1, a_2, \dots, a_n$)

Constant

The fixed value on the right-hand side (b)

System

Two or more equations sharing the same set of unknowns

General Form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

Simple 2-Variable System

Equation	Coefficients	Variables	Constant
$3x + 2y = 12$	3, 2	x, y	12
$x + 4y = 10$	1, 4	x, y	10

□ Each equation is one piece of information. n unknowns generally need n equations to find a unique solution.

Solving by Substitution

Given System

$$x + y = 7 \quad \dots (1)$$

$$x - y = 1 \quad \dots (2)$$

✓ Final Answer

$$x = 4$$

$$y = 3$$

Step-by-Step Solution

1 Isolate x from equation (1)

$$x = 7 - y$$

2 Substitute into equation (2)

$$(7 - y) - y = 1$$

3 Simplify

$$7 - 2y = 1 \rightarrow 2y = 6 \rightarrow \mathbf{y = 3}$$

4 Back-substitute

$$x = 7 - 3 = \mathbf{4}$$

□ Express one variable in terms of the other, then reduce to a single-variable equation — easy to solve directly.

Solving by Elimination

Given System

$$2x + y = 8 \quad \dots (1)$$

$$x + y = 5 \quad \dots (2)$$

✓ Final Answer

$$x = 3$$

$$y = 2$$

Step-by-Step Solution

1 Subtract equation (2) from (1)

$$(2x + y) - (x + y) = 8 - 5$$

2 Solve for x

$$x = 3$$

3 Substitute $x = 3$ into equation (2)

$$3 + y = 5 \rightarrow y = 2$$

□ Add or subtract equations to cancel out one variable entirely — leaving a simpler equation with only one unknown.

When to Use Each Method

Substitution

One variable is already isolated or easy to isolate

Elimination

Coefficients of one variable are equal or easily made equal

Geometric Interpretation

Each linear equation in 2 variables represents a **straight line** on the xy-plane. Solving the system = finding where the lines meet.

Case 1 — Unique Solution

Lines intersect at **exactly one point**

One (x, y) satisfies both equations

$\det(A) \neq 0$

Case 2 — No Solution

Lines are **parallel** (same slope, different intercept)

They never meet — no common point exists

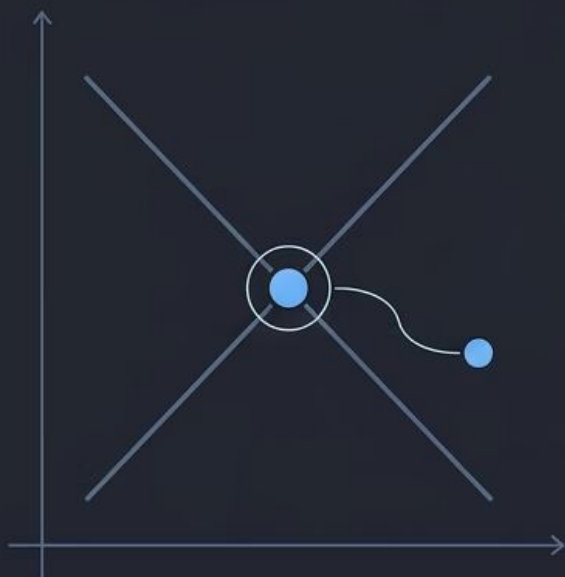
$\det(A) = 0$, inconsistent B

Case 3 — Infinitely Many Solutions

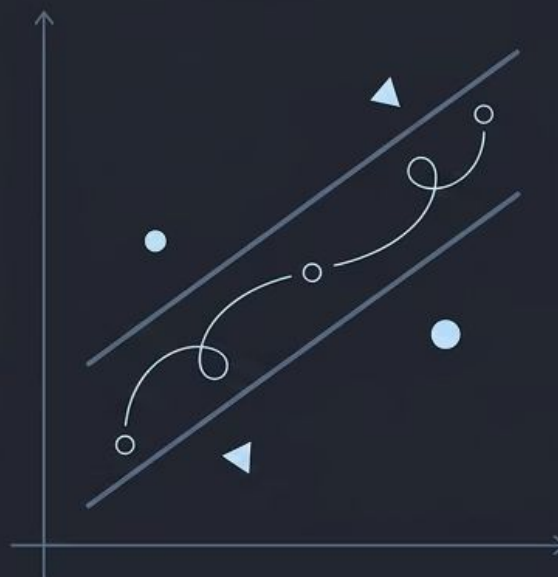
Lines are **identical** (same slope, same intercept)

Every point on the line is a solution

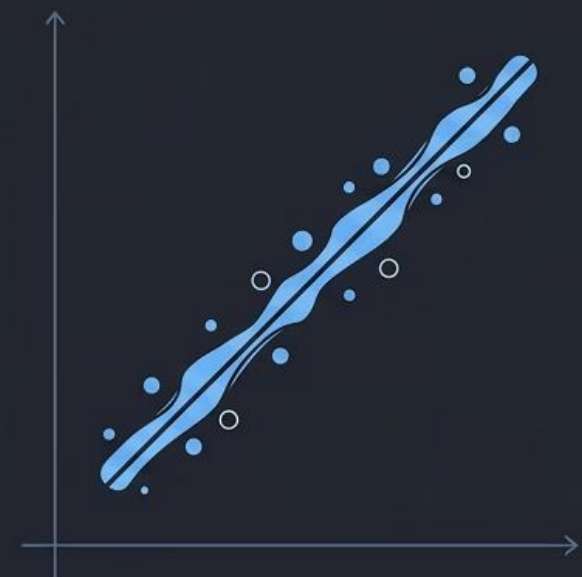
$\det(A) = 0$, consistent B



**Unique
Unique
Solution — one
intersection**



**No Solution —
parallel lines**



**Infinitely
Many
Solutions —
same line**

□ In 3D, each equation is a plane. Three planes can meet at a point, along a line, or not at all — same three cases.

Matrix Form of a System: $AX = B$

Starting System

$$2x + 3y = 8$$

$$x + y = 5$$

Converting to Matrix Form

Coefficient Matrix A

$[2, 3], [1, 1]$ ← collect all coefficients row by row

Variable Vector X

$[x, y]^T$ ← the unknowns we want to find

Constant Vector B

$[8, 5]^T$ ← the right-hand side values

The Matrix Equation: $A \cdot X = B$

$$\begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \end{bmatrix}$$

Why This Matters

- Compact notation — works for 2 equations or 2000 equations
- Enables powerful tools: matrix inverse, determinants, decompositions
- Directly used in ML: linear regression writes the normal equations as $A^T A X = A^T B$

□ $AX = B$ is the matrix version of $ax = b$. Just as $x = b/a$, we seek $X = A^{-1}B$ — if A is invertible.

Determinants: The Basic Idea

What Is a Determinant?

A **determinant** is a single scalar number computed from a **square matrix**. Written as $\det(A)$ or $|A|$.

Formula for a 2×2 Matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\det(A) = ad - bc$$

Multiply along the main diagonal () , subtract the product along the anti-diagonal () .

What Does the Determinant Tell Us?

Invertibility

$\det(A) \neq 0 \rightarrow$ matrix is invertible; $\det(A) = 0 \rightarrow$ matrix is singular (not invertible)

Unique Solution

$AX = B$ has a unique solution if and only if $\det(A) \neq 0$

Geometric Scaling

$|\det(A)|$ measures how much the transformation scales areas (2D) or volumes (3D)

Information Collapse

$\det(A) = 0$ means rows/columns are linearly dependent — the matrix "squashes" space into a lower dimension

💡 Think of $\det(A)$ as a "health check" on the matrix. Non-zero = healthy and invertible. Zero = degenerate, information lost.

Example: Determinant of a 2×2 Matrix

Given Matrix

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$$

Step-by-Step Calculation

1 Identify elements

$$a = 3, b = 2, c = 1, d = 4$$

2 Apply formula

$$\det(A) = ad - bc$$

3 Substitute values

$$\det(A) = (3)(4) - (2)(1) = 12 - 2$$

4 Result

$$\det(A) = 10$$

✓ Interpretation

- $\det(A) = 10 \neq 0 \rightarrow$ **Matrix A is invertible**
 - The system $AX = B$ has a **unique solution**
 - The transformation represented by A scales areas by a factor of **10**
 - We can safely compute $X = A^{-1}B$
- 📌 A non-zero determinant confirms that the two rows of A carry independent information — neither equation is redundant.

Determinant and Solvability

$$\det(A) \neq 0$$

Unique solution exists

Matrix is **invertible**

Rows are **linearly independent**

$$\det(A) = 0$$

No unique solution

Matrix is **singular**

Rows are **linearly dependent**

Example of a Singular Matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$\det(A) = (1)(4) - (2)(2) = 4 - 4 = 0$$

Why Is $\det(A) = 0$ Here?

- Row 2 = $2 \times$ Row 1 $\rightarrow$ the rows are **linearly dependent**
- The second equation gives no new information beyond the first
- Geometrically: both equations represent the **same line** (or parallel lines)

Two Sub-Cases When $\det(A) = 0$

Infinitely Many Solutions

B is consistent with the dependent equations (same line)

No Solution

B is inconsistent (parallel lines that never meet)

📌 A zero determinant means the matrix "collapses" space — it cannot be reversed, so we cannot uniquely recover X from B .

Cramer's Rule

What Is Cramer's Rule?

A formula to solve a square linear system using **determinants only**.
Useful for small systems and theoretical analysis.

For a 2×2 System

$$ax + by = e$$

$$cx + dy = f$$

⚠️ ⚠️ Cramer's Rule only works when $D \neq 0$ (unique solution exists).

Three-Step Process

1 Compute the Main Determinant

$$D = \det(A) = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

2 Compute Numerator Determinants

$$D_x: \text{Replace column 1 with B: } D_x = \begin{vmatrix} e & b \\ f & d \end{vmatrix} = ed - bf$$

$$D_y: \text{Replace column 2 with B: } D_y = \begin{vmatrix} a & e \\ c & f \end{vmatrix} = af - ec$$

3 Solve

$$x = D_x / D \quad y = D_y / D$$

📌 Each unknown is solved by "swapping in" the constants B into the corresponding column of A, then dividing by the main determinant.

Worked Example: Cramer's Rule

System

$$2x + y = 7 \quad \dots (1)$$

$$x + 3y = 11 \quad \dots (2)$$

Step 1 — Main Determinant D

$$D = \begin{vmatrix} 2 & 1 \\ 1 & 3 \end{vmatrix} = (2)(3) - (1)(1) = 6 - 1 = \mathbf{5}$$

$D \neq 0 \rightarrow$ unique solution exists ✓

Step 2 — D_x (replace column 1 with B)

$$D_x = \begin{vmatrix} 7 & 1 \\ 11 & 3 \end{vmatrix} = (7)(3) - (1)(11) = 21 - 11 = \mathbf{10}$$

Step 3 — D_y (replace column 2 with B)

$$D_y = \begin{vmatrix} 2 & 7 \\ 1 & 11 \end{vmatrix} = (2)(11) - (7)(1) = 22 - 7 = \mathbf{15}$$

Step 4 — Solve

$$x = D_x / D = 10 / 5 = \mathbf{2}$$

$$y = D_y / D = 15 / 5 = \mathbf{3}$$

✓ Final Answer

$$x = 2$$

$$y = 3$$

Verification

Equation (1)

$$2(2) + 3 = 7 \quad \checkmark$$

Equation (2)

$$2 + 3(3) = 11 \quad \checkmark$$

Determinant and Geometry

The Geometric Meaning

A matrix A represents a **linear transformation** — it stretches, rotates, or shears space. The determinant measures **how much area (2D) or volume (3D) changes**.

Area Scaling in 2D

$$|\det(A)| = 2$$

Area doubles after transformation

$$|\det(A)| = 0.5$$

Area shrinks to half

$$|\det(A)| = 1$$

Area is preserved (e.g., rotation matrices)

$$|\det(A)| = 0$$

Area collapses to zero — shape squashed onto a line or point

Sign of the Determinant

$$\det(A) > 0$$

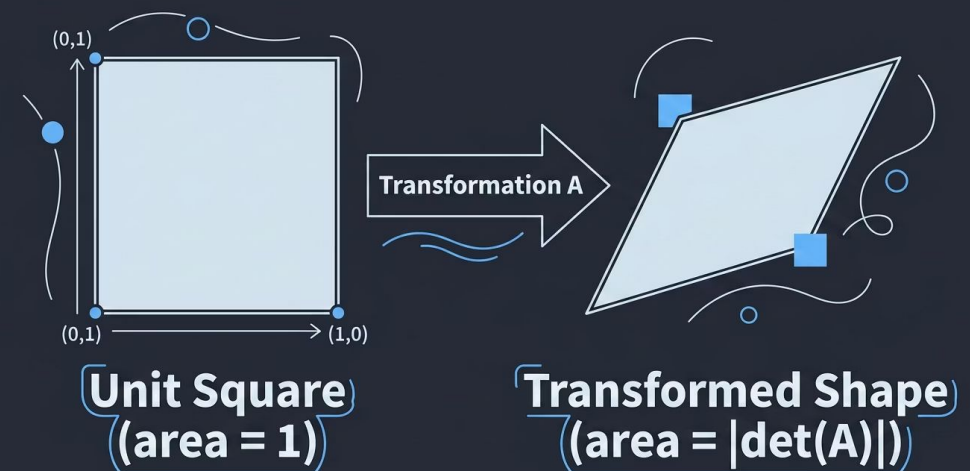
Orientation is **preserved** (no flip)

$$\det(A) < 0$$

Orientation is **reversed** (reflection/flip)

Extension to 3D

For a 3×3 matrix: $|\det(A)|$ = volume scaling factor. A unit cube transforms into a parallelepiped with volume = $|\det(A)|$.



- The determinant is the "zoom factor" of the transformation. Zero means total collapse — irreversible.

Connection to AI/ML + Key Takeaways

Where This Appears in AI/ML



Linear Regression

Solving $A^T A X = A^T B$ for optimal model weights



Least Squares

Minimizing error $\rightarrow$ reduces to a linear system



Neural Networks

Each layer computes $Wx + b$ — a linear transformation



Feature Transformations

PCA, whitening, and normalization use matrix operations



Matrix Inversion

Required when $\det(A) \neq 0$; used in closed-form solutions



Optimization

Gradient conditions often yield linear systems to solve

Conceptual Pipeline

Linear Equations

Matrix Form $AX=B$

Check $\det(A)$

Solve for X

Key Takeaways

01

A system contains multiple equations sharing the same unknowns

02

Any linear system can be written compactly as **$AX = B$**

03

The determinant is defined only for **square matrices**

04

$\det(A) \neq 0$ $\rightarrow$ A is invertible $\rightarrow$ unique solution exists

05

$\det(A) = 0$ $\rightarrow$ A is singular $\rightarrow$ no unique solution

06

Determinants measure **geometric scaling** and encode **solvability**